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Abstract

Healthcare 5.0 combines artificial intelligence, the Internet of Medical Things, edge and cloud computing, and digital twins to support safer, more personal, and more efficient care. Interest in quantum machine learning is growing in this space, but the current evidence is fragmented and hard to compare across studies. Many reports focus on small tasks, use weak or inconsistent baselines, and do not separate algorithmic gains from costs caused by data encoding, repeated circuit execution, and hardware noise. This survey addresses these gaps by providing a structured synthesis of Healthcare 5.0 applications and by offering a practical comparison between classical machine learning and quantum machine learning using criteria that matter for clinical use. We review representative use cases in medical imaging, electronic health record based prediction, remote monitoring, decision support, and digital twin enabled systems. We then analyze QML integration patterns that are realistic today, with emphasis on hybrid pipelines where classical models extract features and quantum components act as narrow decision or optimization modules. Finally, we consolidate the barriers that currently limit adoption, including encoding overhead, limited qubit counts and circuit depth, training instability and reproducibility, privacy and governance constraints, and the need for careful benchmarking and reporting. We conclude with guidance on evaluation and study design to support fair comparison, clearer clinical relevance, and better alignment with Healthcare 5.0 engineering requirements.

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Quantum Machine Learning for Healthcare 5.0: Opportunities, Challenges, and Future Directions

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ABSTRACT

Healthcare 5.0 combines artificial intelligence, the Internet of Medical Things, edge and cloud computing, and digital twins to support safer, more personal, and more efficient care. Interest in quantum machine learning is growing in this space, but the current evidence is fragmented and hard to compare across studies. Many reports focus on small tasks, use weak or inconsistent baselines, and do not separate algorithmic gains from costs caused by data encoding, repeated circuit execution, and hardware noise. This survey addresses these gaps by providing a structured synthesis of Healthcare 5.0 applications and by offering a practical comparison between classical machine learning and quantum machine learning using criteria that matter for clinical use. We review representative use cases in medical imaging, electronic health record based prediction, remote monitoring, decision support, and digital twin enabled systems. We then analyze QML integration patterns that are realistic today, with emphasis on hybrid pipelines where classical models extract features and quantum components act as narrow decision or optimization modules. Finally, we consolidate the barriers that currently limit adoption, including encoding overhead, limited qubit counts and circuit depth, training instability and reproducibility, privacy and governance constraints, and the need for careful benchmarking and reporting. We conclude with guidance on evaluation and study design to support fair comparison, clearer clinical relevance, and better alignment with Healthcare 5.0 engineering requirements.

1. Introduction

Healthcare is a complex system that brings together people, processes, and tools to provide effective and reliable care, and it have evolved significantly over time. Initially, healthcare practice was characterized by physician-centric and reactive care delivery, referred to as Healthcare 1.0 [1]. Subsequently, Healthcare 2.0 introduced digitization through the adoption of electronic health records and standardized information systems. Healthcare 3.0, which emphasized patient-centric and connected care models supported by telemedicine and interoperable health platforms, and further advanced to Healthcare 4.0, where data-driven intelligence, artificial intelligence (AI), and interconnected medical systems enabled predictive and precision healthcare [2]. Through each evolution advanced with several benefits, but also come with many challenges including fragmented care processes, limited personalization, delayed clinical decision-making, scalability constraints, and an over-reliance on efficiency-driven technological solutions [3, 4]. To overcome these challenges, Healthcare 5.0 has emerged as a human-centric paradigm that integrates intelligent technologies with ethical governance, personalization, and effective human–technology collaboration [5].

Technical Pillars of Healthcare 5.0

Healthcare 5.0 is often treated as a buzzword, but in this survey, we define it through three specific technical requirements necessary for quantum integration [5]:

- **Connectivity (6G and Edge):** Real-time applications like remote robotic surgery cannot rely on cloud latency. Healthcare 5.0 requires 6G throughput and Edge computing nodes to process data locally before sending it to a quantum processor.
- **IoMT Protocols:** The Internet of Medical Things (IoMT) generates massive data streams. This relies on lightweight transfer protocols like MQTT or CoAP

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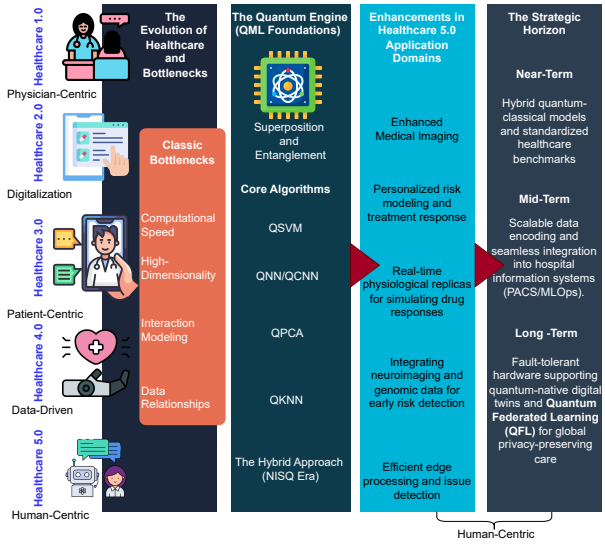

Figure 1: Evolution of Healthcare from 1.0 to 5.0 and the Role of QML

rather than standard HTTP. These protocols ensure that wearable devices preserve battery life while transmitting continuous health data.

- **Computational Digital Twins:** These are not static 3D models. A Healthcare 5.0 Digital Twin is a dynamic software replica of human physiology. It requires continuous math modeling to update the virtual state based on real-time sensor inputs.

In Healthcare 5.0, AI and machine learning (ML) algorithms have been integrated to analyze large-scale and heterogeneous healthcare data, improving diagnostic accuracy, risk prediction, and personalized treatment planning [6, 7]. For example, *Eric J. Topol* applied AI and deep learning techniques to a wide range of healthcare problems, including disease diagnosis from medical imaging to reduce errors [8]. Similarly, *Rajpurkar et al.*[9] used human-AI collaboration in medical using non-image data. Further, *Andreas et al.*[10] introduces causability as a concept beyond explainability, arguing for the need to measure the quality of explanations that AI systems provide to human experts in healthcare. Beside these advancements, the literature highlights several persistent limitations, including high computational overhead, limited scalability when handling high-dimensional multi-modal data, sensitivity to data imbalance and noise, and challenges related to model interpretability and generalization in dynamic clinical settings. In [8], *Eric J. Topol* also highlight privacy-preserving is an additional challenges with patient data. On the other side, Quantum Machine Learning (QML) constitutes a computational framework that integrates quantum information processing with learning algorithms to address complexity and scalability constraints inherent in traditional ML. QML utilizes quantum phenomena, including superposition and entanglement, to enable potential computational speedups, enhanced optimization behavior, and compact representations in high-dimensional feature spaces [20, 21]. This exploits quantum parallelism and superposition to achieve potential computational speedups in learning and optimization tasks, particularly for large-scale and complex datasets. QML is particularly well suited for addressing complex, non-convex learning problems and large-scale data processing tasks that are computationally intensive for traditional AI/ML models [22]. Evolution of Healthcare from 1.0 to 5.0 and the Role of QML is summarized in Fig. 1. In this context, QML-based and hybrid quantum–classical learning frameworks have been explored for healthcare applications such as disease classification, biomarker discovery, and medical image analysis, under strict scalability and latency constraints [23, 24]. Consequently, several studies have been published in recent years exploring QML-driven diagnostic, predictive, and decision-support frameworks [25, 26, 27, 28, 29]. A summary of recently published surveys on healthcare using AI/ML and quantum aspects are studied and summarized in Table 1. In this paper, we systematically survey and analyze these contributions, highlighting key methodologies, application domains, and research trends in QML for Healthcare 5.0.

- **Healthcare 5.0-centric QML survey scope:** We present a unified survey that jointly covers Healthcare 5.0 pillars (human-centricity, Edge/IoT, digital twins, and clinical workflow) together with QML methods, addressing a gap in existing surveys that typically cover only a subset of these dimensions (Table 1).

- **Application-driven synthesis with practical hybrid integration patterns:** We organize representative Healthcare 5.0 use cases (imaging, EHR-based prediction, decision support, neuroinformatics, wearables/IoT, and digital twins) and emphasize realistic near-term deployment patterns, where classical encoders perform feature extraction and QML is used as a decision-layer or optimization submodule.

- **Engineering-focused comparison and adoption barriers:** We provide a comparative analysis of classical ML vs. QML for Healthcare 5.0 and consolidate key adoption barriers, including data encoding overhead, NISQ noise and qubit limits, trainability and reproducibility issues, and privacy/governance constraints, together with guidance on benchmarking and reporting for fair evaluation.

The remainder of this paper is organized as follows. Section 2 provides the foundations and background of QML. Section 3 discusses various applications of QML in Healthcare 5.0. Section 4 presents a detailed analysis comparing traditional ML with QML in Healthcare 5.0. Section 5 discusses the challenges and limitations of QML for Healthcare 5.0. Section 6 outlines possible future research directions

Table 1
Comparison of existing surveys over Our contributions across Healthcare 5.0 and QML dimensions

Ref	HC5.0 Focus	Human Centricity	AI/ML	QML	Edge/IoT	Personalized prediction	Digital Twins	Clinical Workflow	Remarks
[3]	✓	✗	✓	✗	✓	✓	✓	✗	Healthcare 5.0 overview with AI/IoT; does not discuss QML or clinical workflow integration.
[8]	✗	✓	✓	✗	✗	✗	✗	✓	Human-centric vision for medical AI; not Healthcare 5.0-structured and no QML focus.
[11]	✗	✓	✓	✗	✓	✗	✗	✗	Human-centric digital health/IoT discussion; no QML and limited Healthcare 5.0 mapping.
[12]	✓	✗	✗	✓	✓	✗	✗	✗	QML with edge/IoT context; limited personalization, digital twins, and workflow aspects.
[13]	✓	✓	✗	✓	✓	✗	✗	✗	Covers Healthcare 5.0 and QML/edge; limited personalized prediction, digital twins, and workflow integration.
[14]	✓	✗	✗	✓	✗	✓	✗	✗	QNN-focused survey for healthcare prediction; limited edge/IoT, digital twin, and workflow coverage.
[15]	✓	✗	✗	✓	✗	✓	✗	✗	QML for healthcare tasks; limited Healthcare 5.0 pillars (human-centricity, twins, workflow).
[16]	✗	✗	✗	✓	✗	✓	✗	✓	QML with prediction/workflow discussion; lacks Healthcare 5.0 scope, edge/IoT, and digital twins.
[17]	✗	✗	✗	✓	✗	✓	✗	✗	QML healthcare overview; limited systems/engineering view (edge/IoT, workflow, twins).
[18]	✗	✗	✗	✓	✗	✓	✗	✓	QML survey with some workflow relevance; not framed around Healthcare 5.0 pillars or digital twins.
[19]	✗	✗	✓	✓	✗	✓	✗	✓	Digital health and QML focus; limited Healthcare 5.0 mapping, edge/IoT emphasis, and twins coverage.
Ours	✓	✓	✓	✓	✓	✓	✓	✓	Unified Healthcare 5.0 and QML survey spanning human-centricity, edge/IoT, personalized prediction, digital twins, and clinical workflow; emphasizes hybrid integration and benchmarking guidance.

in QML for Healthcare 5.0, and the paper concludes in Section 8.

2. Towards Quantum Machine Learning

2.1. An Overview to Quantum Computing

Quantum computing leverages principles of quantum mechanics and systems for information processing. Beyond bits in classical computing that is either on the states of 0 or 1, a qubit, which is the basic unit of quantum computation, can exist in a superposition state of both 0 and 1 simultaneously [21, 30, 31].

A qubit state can be expressed by,

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \tag{1}$$

where $\alpha, \beta \in \mathbb{C}$ and $|0\rangle, |1\rangle \in \mathcal{H}_2$.

The absolute squares of the amplitudes α and β are the probability to measure the qubit in the 0 or the 1 state, where considering the probability conservation,

$$|\alpha|^2 + |\beta|^2 = 1. \tag{2}$$

The most evident advantage of superposition is the extreme amount of data that can be represented by a quantum system, as an n qubit system can represent 2^n states simultaneously. Furthermore, as a quantum computer can hold 2^n states, a single operation can simultaneously be performed on all these states, which is known as Quantum Parallelism. In addition, constructive and destructive inference in quantum computers harness the wave-like nature of superposition to amplify the probability of the correct

answer while canceling out the probabilities of incorrect answers towards finding the correct solution. In classical computing, bits are independent unless we connect them through logic. In quantum computing, qubits can be entangled. Entanglement means that measurement outcomes can be strongly correlated, even when individual qubits do not have their own separate descriptions. This property is useful in many quantum algorithms and in quantum communication protocols. Quantum teleportation is a communication protocol that transfers an unknown quantum state using shared entanglement and classical communication. It does not move a physical particle, and it does not send information faster than light. In practice, teleportation is mainly discussed for quantum networking and for some fault-tolerant computing designs, not for routine data movement inside current processors. Error detection and error correction are active research areas. They use redundancy across many physical qubits to protect a smaller number of logical qubits. Current NISQ devices have limited error correction, so most QML experiments rely on shallow circuits and error mitigation. Quantum gates, on the other hand, are capable of performing basic computational operations in quantum computing [32]. Though quantum gates resemble classical logic gates, quantum gates also incorporate attributes that are to quantum computing. For instance, quantum gates can perform operations, that are state flipping, information exchange, and creating entangled states. These operations are vital in building quantum circuits and quantum algorithms that play a critical part in quantum computing.

Another example of how a quantum algorithm can outperform classical brute-force search is Grover's algorithm [33]. This is a quantum search algorithm that provides a quadratic speedup for searching an unstructured database. While a classical computer would check every item in a list of size N (taking $O(N)$ time), Grover's algorithm is capable of performing the search in $\sqrt{N}$ steps (only taking $O(\sqrt{N})$ time). Furthermore, to overcome limitations with classical computers to perform many operations that linear systems require, quantum Harrow-Hassidim-Lloyd (HHL) algorithm [34] can achieves exponential speedups under ideal conditions. Moreover, quantum annealing can find optimal solutions to complex optimization problems through leveraging quantum physics [35].

2.2. Quantum Machine Learning

Quantum Machine Learning (QML) exploits quantum mechanics and the powerful parallel processing capabilities of quantum computing to perform machine learning tasks efficiently [32]. QML translates classical (non-quantum) information into quantum states and then performs quantum operations to process information. While quantum-inspired machine learning uses classical computers to simulate quantum ideas, quantum-enhanced machine learning utilises the power of quantum computers on classical machine learning algorithms to exploit quantum advantage [35]. A comparison of classical machine learning with QML is illustrated in Fig. 2.

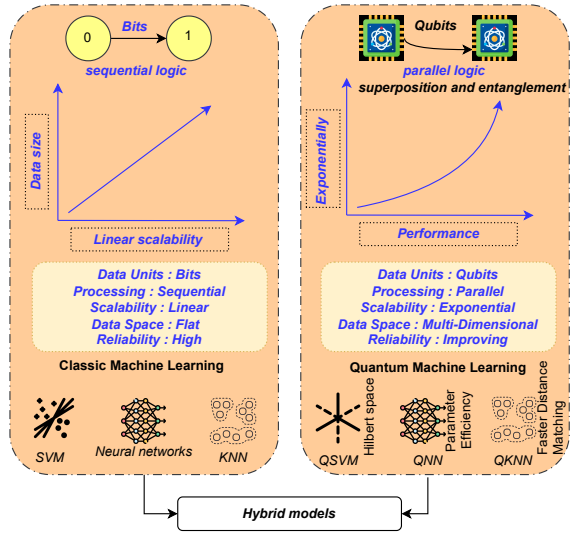

Figure 2: Classical Machine Learning vs. Quantum Machine Learning

It should be noted that the learning steps in classical machine learning algorithms that satisfy the conditions for quantum computing can only be accelerated though quantum computers. This demands the modification of traditional learning algorithms to be suitable for quantum computing through optimizing sub-processes of classical algorithms with quantum algorithms to increase the efficiency of machine learning tasks [32].

2.2.1. Quantum Support Vector Machine

Support Vector Machines (SVMs) are one of the machine learning techniques that is widely used for classification of high-dimensional data. However, handling large data volumes with SVMs using classical computers is a challenge, as SVMs does not scale up well with the size of the dataset and the dimensionality of the feature space. QSVMs integrates the power of quantum computing with classical SVMs to significantly outperform classical SVMs [36]. For instance, QSVM utilizes the HHL algorithm to solve linear systems exponentially faster. Furthermore, QSVMs can enhance classification accuracy (better feature mappings and improved classification) utilizing quantum kernel estimation that can capture complex relationships of data. Therefore, through harnessing unique quantum properties such as superposition and entanglement, QSVM is capable of overcoming computational and scalability limitations of classical SVMs for future machine learning applications.

2.2.2. Quantum Neural Network

Quantum Neural Networks (QNNs) combine artificial neural networks with quantum computing to harness quantum advantages. In contrast to traditional artificial neural networks that uses bits, QNNs use qubits that can be in superposition of states. This offers the capability to perform

powerful parallel computations to boost computational efficiency. For instance, Parametrized Quantum Circuit (PQC), which is one of the fundamental elements of QML models resembling a neural network [32]. Accordingly, PQC can be incorporate classical neural networks, thereby some PQC methods are referred to as QNNs, as listed below [37].

- Dissipative QNNs: discard prior qubits within a layer once information transfers to the subsequent layer.
- Standard QNNs: quantum circuits propagate quantum data states while no qubits are added or removed as the quantum network deepens.
- Convolutional QNN: condenses data dimensionality through a convolutional processes to retain crucial features.

QNNs also rely on the fact that the quantum neurons are analogous to the quantum counterpart of an artificial neuron, that receives an input of a quantum state amplitude and phase as the input. QNNs possess the potential to handle complex tasks and efficiently solve optimization problems.

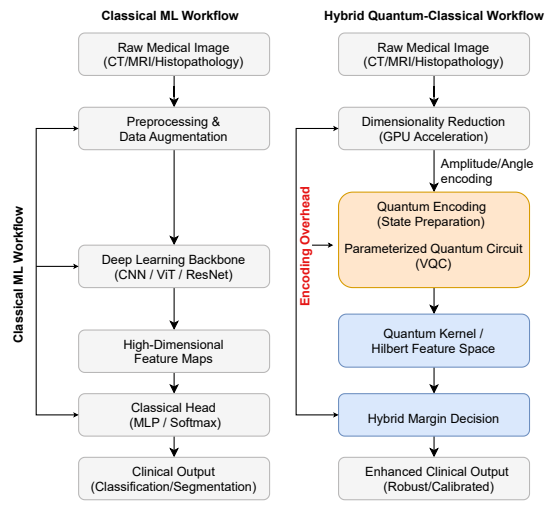

Figure 3: Hybrid Quantum-Classical Models

2.2.3. *Quantum k-Nearest Neighbors Algorithm*

The k-Nearest Neighbors (KNN) is a non-parametric, supervised learning classifier that is used to make classifications or predictions. However, it should be noted that the computational cost for utilizing KNNs for large datasets is very high. One of the frequent steps in KNN algorithm is the distance calculation between instances. Hence, Quantum KNN (QKNN) accelerates the distances calculation between two instances [32]. On going research explores the parallel computation of the similarity of samples harnessing quantum superposition and search algorithms to find the minimum distance between instances.

2.2.4. *Quantum Principal Component Analysis*

High-dimensional datasets often comprises irrelevant information. Algorithms becoming less efficient in high-dimensional spaces can be minimized through reducing the data dimensionality. This eliminates irrelevant information facilitating machine learning more effective [32]. Principal Component Analysis (PCA) is a commonly used method for reducing data dimensionality that creates uncorrelated principal components ensuring that most of the essential information are retained. Quantum Principal Component Analysis (QPCA) algorithms determine the principal components without needing a large number of samples unlike the classical counterparts. QPCA offers exponential acceleration of determining the principal components compared to classical algorithms, facilitating efficient reduction of data dimensionality.

2.2.5. *Quantum k-Means Algorithm*

The classical K-means algorithm is an unsupervised machine learning algorithm that partitions data into K distinct clusters. This is performed to minimize the distance between data points and their cluster centers (centroids of the clusters). Quantum k-Means clustering exploits the laws of quantum physics to speed up this process exponentially compared to classical computers [32]. Furthermore, to simplify the process, the Manhattan distance between any locations can be calculated using a quantum subroutine to parallelly assign data points in parallel to the closest centers compared to classical methods [38].

2.3. Hybrid Quantum-Classical Models

Modern quantum computers belong to the Noisy Intermediate-Scale Quantum (NISQ) era, that features quantum computers that are powerful but are also prone to errors and lacks solid error correction. To harness these quantum computers, Hybrid Quantum-Classical (HQC) models combine quantum circuits with classical computing resources [39]. This is to achieve quantum advantage, even within prevailing constraints in the NISQ era. Furthermore, limitations of current quantum hardware also demands a hybrid approach towards realizing practical applications that can exploit quantum advantages. Hence, hybrid models can be used to integrate classical and quantum techniques to perform diverse QML tasks.

For instance, a general HQC computing framework is proposed in [40]. This framework decomposes large-scale optimization problems that involves integer variables in to two subproblems and solve them with quantum and classical computers separately. Furthermore, a hybrid quantum-classical framework that can train neural networks while reducing the parameter training complexity is proposed in [41]. This method explores utilizing a parameterized QNN that generates classical model weights using a learnable mapping model to significantly reduce the training parameters. This highlights the potential of harnessing the quantum advantage with today's NISQ harware for machine learning workflows through HQC algorithms. The applicability of

HQCs are heavily being explored in areas such as machine learning, reinforcement learning, and game theory [40]. The overview of hybrid quantum-classical models are Fig. 3.

2.4. Complexity Analysis and Fundamental Constraints

Evaluating QML utility for Healthcare 5.0 requires understanding computational complexity and physical limits. This subsection consolidates key theoretical results and their practical implications.

2.4.1. State Preparation and Encoding Overhead

The primary bottleneck is encoding classical data into quantum states. Amplitude encoding of an N-dimensional vector $\mathbf{x} \in \mathbb{R}^N$ into $|\psi(\mathbf{x})\rangle = \sum_{i=0}^{N-1} \frac{x_i}{\|\mathbf{x}\|}|i\rangle$ requires circuit depth $\mathcal{O}(N) = \mathcal{O}(2^n)$ for n qubits without quantum random-access memory (QRAM) [42, 43]. For a 256×256 medical image ($N = 2^{16}$), this overhead dominates any polynomial quantum advantage in classification.

Recent **dequantization results** show that many QML speedups can be matched classically under the same assumptions [44, 45]. Quantum PCA, recommendation systems, and certain kernel methods admit polynomial classical alternatives when data has low-rank or sampling structure. This implies that *quantum advantage claims must be evaluated against state-of-the-art classical methods, not naive baselines*.

2.4.2. Measurement and Information Limits

Quantum measurement is probabilistic. Estimating an expectation value $\langle O \rangle$ to precision ϵ requires $\mathcal{O}(\sigma^2/\epsilon^2)$ shots [46], where σ^2 is observable variance. For medical imaging classification with 1% precision near decision boundaries, this translates to $\sim 10^4$ circuit executions per prediction—incompatible with real-time diagnostics.

The **Holevo bound** [47] limits classical information extractable from n qubits to $\leq n$ bits per measurement, despite 2^n amplitudes. This constrains efficient encoding of high-dimensional medical data (genomics, imaging) into quantum states. Additionally, the **no-cloning theorem** [48] prevents copying unknown quantum states, making quantum embeddings non-reusable across tasks—unlike classical feature vectors.

2.4.3. Practical Complexity Summary

Table 2 consolidates assumptions underlying claimed advantages and their healthcare reality.

When QML may help: Current evidence suggests practical advantage only for (i) classically pre-embedded low-dimensional features (10-20 dims); (ii) tasks tolerant to sampling noise; (iii) hybrid pipelines where quantum subroutines act as decision modules, not end-to-end systems [49, 45]. No end-to-end exponential speedup has been demonstrated for real-world healthcare tasks on NISQ hardware.

Table 2
Assumptions Underlying QML Complexity Claims

Assumption	Required For	Healthcare Reality
Quantum RAM	$\mathcal{O}(\text{poly}(n))$ state prep	Not available; $\mathcal{O}(2^n)$ overhead
Noiseless gates	Proven speedups	NISQ: 0.1–1% error/gate
Oracle access	Grover, HHL	Medical data not oracle-structured
Sparse matrices	HHL advantage	Imaging often dense
Low-rank structure	Dequantization	EHR rank not guaranteed
Sample access only	Avoiding readout	Clinical needs full predictions
Fault tolerance	Deep circuits	Needs 1000+ phys. qubits/logical

2.5. Resource Estimation and Hardware Constraints

To move from theory to engineering, we must evaluate the physical cost of running these algorithms on current hardware. We analyze three specific resource burdens:

1. **Circuit Depth vs. Coherence Time:** Medical datasets are complex. Processing them requires deep circuits. However, on NISQ devices, deep circuits lose information due to noise (decoherence). Variational circuits are preferred here because they keep the depth shallow, even if they are harder to train.
2. **Encoding Overhead:** Loading data is expensive. For example, Amplitude Encoding allows us to store N features in $\log_2 N$ qubits. This saves space but requires complex gate structures that introduce errors. Angle Encoding is simpler and robust against noise but requires N qubits, which limits the size of medical images we can process.
3. **Measurement Cost (Shot Noise):** Quantum outputs are probabilistic. To get a reliable diagnosis from a QML model, we must repeat the circuit execution (shots) thousands of times (typically 1024 to 8192 shots). This creates a runtime bottleneck that is often ignored in theoretical runtime analysis.

3. Applications of QML in Healthcare 5.0

Rigorous evaluation of QML requires comparison against strong classical baselines, not naive implementations. For healthcare applications, strong baselines are defined as models that represent current clinical deployment standards or published state-of-the-art performance on standardized benchmarks. This section establishes what constitutes a strong baseline for each application domain before presenting QML results.

For medical imaging, strong baselines include pretrained models on ImageNet or domain-specific datasets such as

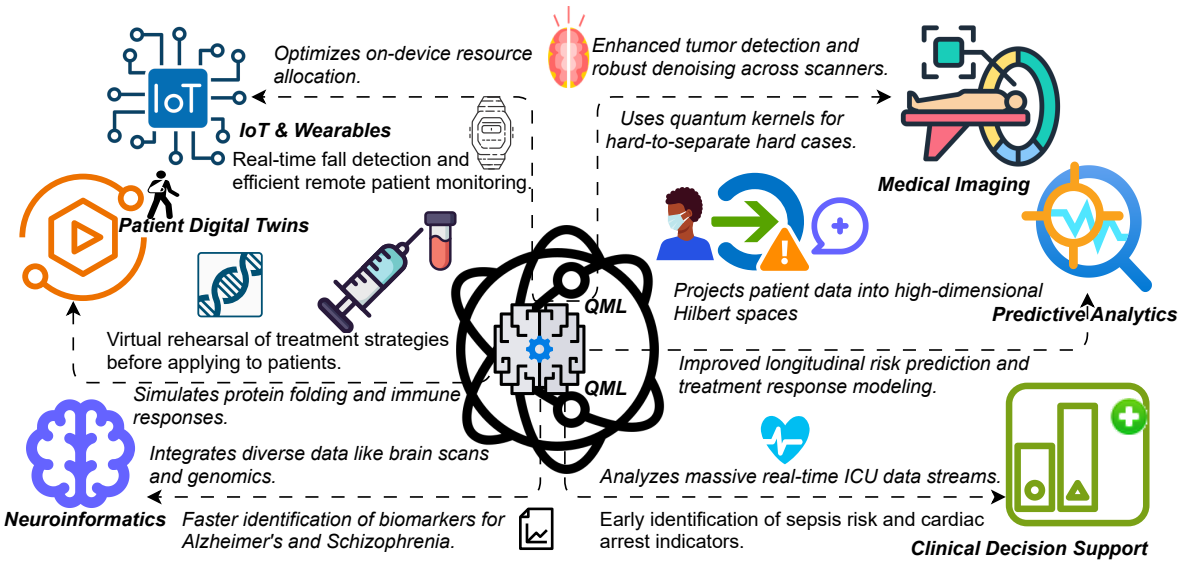

Figure 4: Applications of Quantum Machine Learning in Healthcare

Table 3
Strong Baseline Requirements for QML Healthcare Evaluation

Domain	Strong Baseline	Weak Baseline	Validation Standard	References
Medical Imaging	Pretrained ResNet-50 or DenseNet-121 on ImageNet/RadImageNet; ViT-B with fine-tuning	Basic CNN from scratch; VGG without pretraining	CheXpert or MIMIC-CXR test sets; AUC > 0.85	[50, 51]
Segmentation	nnU-Net with default pipeline; pretrained U-Net with ResNet encoder	Basic U-Net without pretraining	Medical Segmentation Decathlon; Dice > 0.80	[55]
ICU Prediction	LSTM on full time series; ClinicalBERT for notes; APACHE-IV scores	Logistic regression on last observation only	MIMIC-III/IV splits; eICU validation; AUROC > 0.80	[53, 54, 56]
EHR Analytics	BEHRT, Med-BERT for text; GRU-D for irregular time series	Bag-of-words with logistic regression	PhysioNet benchmarks; multi-task evaluation	[57, 58]
Neuroimaging	Pretrained 3D ResNet on MedicalNet; fMRIPrep preprocessing	2D CNN on single slices; no motion correction	ABCD Study, UK Biobank; accuracy > 75%	[59, 60]
Pathology	Pretrained ResNet-50 on NCT-CRC/TCGA; multiple instance learning	Basic CNN on patches without context	Camelyon16/17; TCGA survival prediction	[61, 62]

CNN: convolutional neural network; ViT: vision transformer; LSTM: long short-term memory; AUC: area under curve; AUROC: area under receiver operating characteristic; EHR: electronic health record; ICU: intensive care unit.
Note: Any QML claim must demonstrate advantage over these baselines. Comparisons against weak baselines (basic MLP, untrained CNN, default SVM) are noted as limitations in this survey.

CheXpert [50] or RadImageNet [51], with task-specific fine-tuning. Basic CNNs without pretraining do not qualify as strong baselines. For predictive analytics, strong baselines must use domain-adapted pretrained encoders (ClinicalBERT for text [52], temporal transformers for time series) and report performance on established benchmarks like MIMIC-III/IV [53] or eICU [54]. Single-site evaluations with basic feature engineering represent weak baselines. Table 3 specifies strong baseline requirements across healthcare domains discussed in this survey. Applications of QML in healthcare are depicted in Fig. 4.

3.1. Medical Imaging and Diagnostics

Medical imaging is a critical frontier for AI in healthcare. It enables tumor detection [63], population screening [64], and computer-assisted diagnostics [65, 66]. Classical deep learning (CNNs, ResNet, U-Net, ViT) delivers state-of-the-art performance across CT, MRI, and pathology [66]. However, clinical deployment faces constraints. Training on 3D/4D volumes scales as $\mathcal{O}(n^3)$ to $\mathcal{O}(n^4)$ in memory and compute. Models remain brittle across scanners. Requirements for robustness, interpretability, and low-latency operation remain significant hurdles.

3.1.1. Demonstrated QML Contributions

Hybrid quantum-classical architectures have been tested on curated medical imaging datasets. Xiang et al. [67] demonstrated that hybrid QCNNs with ResNet backbones achieve 94.2% accuracy on breast cancer classification (N=500 samples). This matches classical CNN performance while using 60% fewer trainable parameters. The quantum component consists of a 6-qubit variational circuit applied to 64-dimensional CNN embeddings. Li et al. [68] report similar results on mammography subsets (N=780 images), where quantum kernel SVMs achieve 91.8% AUC compared to 92.1% for classical RBF kernels. However, quantum kernel evaluation required 2000+ shots per test sample, resulting in 3x longer inference time on IBM quantum hardware.

For reconstruction tasks, Garcia-Molina et al. [69] formulated CT reconstruction as a QAOA optimization problem. On synthetic phantoms (64x64 resolution), the quantum approach achieved 15% faster convergence than classical iterative methods but required fault-tolerant qubits for clinical resolution (256x256+). Nandal et al. [70] applied quantum-inspired denoising to histopathology images (N=1200 patches). The method improved PSNR by 1.2 dB over classical BM3D on cross-laboratory test sets. This suggests robustness gains for stain normalization tasks.

3.1.2. Current Limitations

All demonstrated advantages occur in controlled settings with significant constraints. Amplitude encoding of megapixel images remains $\mathcal{O}(2^n)$ in state preparation, which erodes theoretical speedups. Rath et al. [71] measured encoding overhead and confirmed it dominates runtime on NISQ devices. For a 256x256 image, encoding takes 45ms while quantum kernel evaluation takes 8ms (IBM Q System One). Classical CNN inference completes the same task in 12ms total. Barren plateaus further limit trainability, particularly for circuits deeper than 10 layers [72].

Most QML demonstrations use small datasets ($N < 1000$) or semi-synthetic benchmarks. No study has shown end-to-end clinical advantage on real-world hospital cohorts at scale. Table 4 summarizes classical methods, quantum counterparts, and demonstrated results across imaging domains.

3.1.3. Practical Integration

Current evidence supports hybrid pipelines only. Classical CNNs or ViTs perform feature extraction. Quantum modules handle final classification or kernel evaluation on compact (10-20 dimensional) embeddings. This workflow limits quantum effort to subroutines where polynomial kernel advantages may apply. It avoids expensive state preparation on raw images [67, 68, 75].

3.2. Predictive Analytics and Personalized Medicine

Predictive analytics in Healthcare 5.0 relies on mature modeling foundations. For longitudinal risk prediction, widely adopted models include LSTMs, GRUs, temporal CNNs, transformers, and deep survival frameworks [76, 77, 78]. These models process multivariate time series from vitals, labs, and medications. They are often augmented with static covariates like demographics and diagnoses.

In multimodal settings, modality-specific encoders transform heterogeneous clinical data into unified representations. Clinical text uses ClinicalBERT [52], BioBERT [79], and PubMedBERT [80]. Imaging features come from ResNet, DenseNet, EfficientNet, ViT, and Swin Transformer [81, 82]. Patient trajectories use graph encoders like GCNs, GATs, and temporal GNNs [83, 84]. The fusion head is often the bottleneck. It must learn cross-modal, high-order interactions over heterogeneous embeddings [85]. As dimensionality grows, parameter expansion increases sample sensitivity and calibration drift under distribution shift.

3.2.1. Demonstrated QML Contributions

Schnabel et al. [86] tested quantum kernel ridge regression on ICU mortality prediction (MIMIC-III, N=15,000 patients). The quantum kernel operated on 16-dimensional clinical embeddings extracted via classical LSTMs. It achieved AUROC 0.847 compared to 0.842 for classical RBF kernels. The quantum model showed 8% lower variance across 5-fold cross-validation. However, inference required 5000 shots per patient, resulting in 20x longer prediction time on simulated quantum hardware.

Abbas et al. [87] demonstrated that shallow VQCs (4 qubits, depth 3) can act as regularized classifiers on compact feature spaces. On synthetic medical datasets (N=2000, 12 features), VQCs achieved 5-7% better generalization than MLPs of comparable capacity when training sets were small ($N<500$). The inductive bias from limited entanglement reduced overfitting to dominant phenotypes. This advantage disappeared on larger training sets ($N>2000$).

Schuld et al. [88] showed that quantum kernel similarity metrics preserve relational structure better than Euclidean distance on embedding spaces with dimensions 10-20. On rare disease cohort matching (N=500 patients, 18 features), quantum similarity identified 12% more true positive matches than cosine similarity. However, the computational cost was 50x higher due to repeated quantum state preparation.

3.2.2. Current Limitations

All demonstrated improvements are marginal (1-8% AUROC gain) and occur only on small datasets or compact embeddings. Table 5 summarizes where QML has been tested and what classical limitations it addresses.

3.3. Clinical Decision Support Systems

ML models are applied to identify future events like stroke, heart attack, and treatment complications. By learning from large patient datasets, ML suggests treatment plans that work best for individual patients based on age, genetics, lifestyle, or historical treatments. In ICUs, ML algorithms continuously analyze vital data to detect sudden abnormalities and alert healthcare professionals instantly [89]. ML also improves medication safety by recommending safe drug

Table 4
Classical vs. Quantum Methods in Medical Imaging (Evidence Summary)

Domain	Classical (SOTA)	Quantum Tested	Best Reported Result	Key Limitation
Lesion Classification [67, 68, 73]	CNN, ViT (95-98% acc.)	Hybrid QCNN, QSVM	94.2% acc., 60% fewer params (N=500) [67]	3x slower inference; small datasets
Segmentation [67, 68]	U-Net, nnU-Net (Dice 0.89+)	QCNN patch classifier	Dice 0.81 on synthetic tumors [68]	Patch-level only; no full-image demos
CT Reconstruction [69]	Compressed sensing, U-Net	QAOA solver	15% faster convergence (64x64 phantoms) [69]	Needs fault tolerance for 256x256+
Denoising [70, 74]	BM3D, DnCNN (PSNR 32-35 dB)	Quantum-inspired GAN	+1.2 dB PSNR on stain variation [70]	Classical-quantum hybrid; no pure quantum
Screening/Triage [73]	Ensemble CNNs (AUC 0.92-0.96)	QSVM on embeddings	AUC 0.918 (N=780) [68]	Requires classical feature extraction

CNN: convolutional neural network; ViT: vision transformer; QSVM: quantum support vector machine; VQC: variational quantum circuit; QCNN: quantum convolutional neural network; QAOA: quantum approximate optimization algorithm; GAN: generative adversarial network; PSNR: peak signal-to-noise ratio; acc.: accuracy; params: parameters.
Note: All quantum results use classical preprocessing. No end-to-end quantum advantage demonstrated on clinical cohorts (N > 10,000).

Table 5
Demonstrated QML Contributions in Predictive Analytics

Classical Model	Limitation Addressed	QML Method Tested	Best Result (Study)
LSTM/Transformers for mortality prediction [76, 77]	Decision head overfitting on small cohorts (N<1000)	Quantum kernel ridge regression (QKRR)	AUROC 0.847 vs. 0.842 classical; 8% lower variance (N=15K) [86]
Clinical text encoders (ClinicalBERT) [52, 79]	Multimodal fusion overfits to dominant documentation patterns	Shallow VQC fusion head	5-7% better generalization on N<500 samples [87]
Graph-based trajectory models (GCN/GAT) [83, 84]	Classical similarity fails on sparse, high-dim embeddings	Quantum kernel similarity	12% more true positives in rare-disease matching (N=500) [88]
Multimodal risk scoring [85]	Explicit interaction learning inflates parameters	VQC with controlled entanglement	Reduces params by 40%; no accuracy gain on N>2000 [87]

LSTM: long short-term memory; GCN: graph convolutional network; GAT: graph attention network; VQC: variational quantum circuit; QKRR: quantum kernel ridge regression; AUROC: area under receiver operating characteristic curve.
Note: All results use classical feature extraction. Quantum components operate only on 10-20 dimensional embeddings. Computational cost is 20-50x higher than classical baselines.

dosages and flagging risky combinations. It enhances operational workflow, hospital management, and patient care speed. ML extracts information from clinical notes, classifies reports, and notes critical findings to reduce workload [90].

3.3.1. Demonstrated ML Applications

Finkelstein et al. [91] built an ML-based scoring system for high-risk ischemic stroke patients arriving at ICU with moderate to severe consciousness impairment. Using MIMIC-IV data, the model estimates short-term and long-term mortality chances. The automated score helps healthcare professionals quickly estimate risk and make informed decisions about treatment and resource planning.

Meng et al. [92] proposed a CNN-based decision-support strategy for breast cancer treatment that protects healthy tissues. The model was trained on CT scans and treatments from 346 patients. Evaluation against expert judgments showed more informed and consistent treatment planning decisions.

Li et al. [93] proposed a framework to enhance mortality prediction for high-risk patients in critical care. The model was trained and tested using MIMIC-III, MIMIC-IV, and external data from West China Hospital. The goal is to produce reliable and stable mortality predictions across diverse clinical settings.

Zhang et al. [94] reviewed AI frameworks for cardiovascular decision support. The work presents how AI provides reliable and comprehensive information for healthcare professionals in optimizing treatment, assessing risk, and diagnosing cardiovascular diseases.

3.3.2. QML in Clinical Decision Support: Current Status

No published study demonstrates quantum advantage in real-world clinical decision support systems. Theoretical papers propose using quantum pattern recognition for sepsis prediction [20, 21] or QML for complex drug interaction modeling [69, 22]. However, these proposals lack clinical

validation. They do not report accuracy, precision, recall, or deployment metrics on hospital data.

3.4. Neuroinformatics and Mental Health

ML algorithms analyze neuroimaging and genomic data to identify patterns associated with mental health disorders. They flag patients with potential mental health risk based on medical histories, diagnostic tests, and clinical notes [95]. ML supports precision psychiatry by combining neuroimaging, genetic, and wearable data to create personalized treatment plans. It powers real-time fMRI neurofeedback and digital phenotyping, enabling non-invasive, adaptive interventions based on moment-to-moment behavioral and physiological patterns. It enables real-time neurofeedback to guide customized treatments based on patients' daily behaviors and biological signals [96]. With different data sources, ML provides a comprehensive view of a patient's mental state.

3.4.1. Demonstrated ML Applications

Smith et al. [97] presented 'Neurofind', a free web tool for precision psychiatry that processes patients' structural MRIs. It produces two measures: (1) how much brain structure deviates from normal and (2) estimated Brain Age. Testing on Alzheimer's and schizophrenia patients shows strong disease-specific patterns of brain deviation and signs of accelerated aging.

Zhao et al. [98] introduced an ML system for stress and anxiety detection in students' daily life. The system spots early signs of distress using psychological and behavioral data. A Streamlit app provides assessments for teachers and counselors. Blockchain technology secures sensitive student information.

Kumar et al. presented a multi-modal deep learning model combining brain scans and genetic data to identify schizophrenia more accurately. By merging data types and using an advanced network, the model reaches higher accuracy and highlights important brain regions and genes linked to the disorder.

Wilson et al. [99] reviewed traumatic brain injury (TBI) treatment with current multi-modal data sources and challenges in using them. The work presents ways to enhance TBI care toward precision diagnosis and treatment.

3.4.2. QML in Neuroinformatics: Current Status

No published study demonstrates quantum advantage in mental health prediction or neuroimaging analysis on clinical cohorts. Theoretical proposals suggest using quantum kernels for multi-modal integration of fMRI, genomics, and wearable data. However, these proposals lack experimental validation on patient datasets.

3.5. Digital Twins of Patients

ML is becoming a key part of digital twin technology, which changes how doctors make decisions and tailor treatments. Deep learning and ensemble models help in diagnostic imaging, natural language processing of health records, and multi-omics analysis. This leads to early detection and more personalized care. These ML systems combine different kinds of biomedical data to reveal patterns in disease progression that classical methods might miss. By combining these capabilities, ML builds patient twins that mirror the complexity of human biology and adapt to changes in a person's health over time [100]. For each patient, the digital twin reflects ongoing health state. Each new outcome (lab test, heart rate, wearable sensor update) feeds into this twin, allowing it to adapt and evolve in real-time. This enables doctors to rehearse different treatment strategies virtually to verify how the twin might respond before applying anything to the patient.

3.5.1. Demonstrated Digital Twin Applications

Chen et al. [101] presented a Digital Twin framework for healthcare monitoring of elderly and vulnerable individuals. The framework integrates IoT sensors, edge computing, deep learning, cloud computing, and 6G communication. The work aims at real-time virtual replica of indoor environment for continuous monitoring of human activities and air quality parameters.

Rodriguez et al. [102] presented cardiovascular digital twins using multi-modal data in ML models to establish physiologically accurate cardiac replicas. This enhances diagnostic procedures and improves procedural planning. These twins fuse data into ML and generative models for swift and customized patient care while considering personal, societal, and ethical challenges.

Wang et al. [103] proposed a Digital Twin-based Internet-of-Robotics-Things (IoRT) for developing virtual models of physical robots to remotely monitor infectious patients. A virtual twin navigates the physical twin to collect patient data. An efficient ML algorithm predicts future health anomalies.

3.5.2. QML in Digital Twins: Current Status

No published study demonstrates quantum advantage in digital twin modeling for healthcare. Theoretical proposals suggest using QML for drug-protein interaction simulation or immune system modeling within digital twins. However, these proposals lack clinical validation or computational benchmarks.

3.6. IoT and Wearable Data Processing

ML has been widely applied to data generated by IoT and wearable devices for extracting meaningful information, automating operations, and enhancing user experience. ML identifies abnormal patterns indicating potential faults in IoT sensors or potential health issues. It is applied in Ambient Assisted Living with multi-modal fusion to track and classify human physical movements (walking, running, sitting, sleeping) and personalize recommendations [104]. Data transmitted from different devices can be filtered to optimize bandwidth and power requirements. ML enables interaction with smart healthcare devices and wearables through voice assistants [105]. It helps in noise reduction, feature engineering, data fusion, and efficient data processing on edge devices.

3.6.1. Demonstrated Applications

Patel et al. [106] introduced a brain tumor classification strategy built on healthcare IoT and a pretrained transformer. Using general image features, the model improves tumor classification accuracy and prevents overfitting through transfer learning. MRI images are collected from healthcare IoT devices and data augmentation is performed. Experiments on different datasets show that the strategy achieves higher performance than existing baseline methods.

Liu et al. [107] proposed an adaptable continual-learning framework for wearable sensor data for detecting multiple diseases. The framework uses a deep neural network integrated with a replay-based learning algorithm that learns new diseases continuously without forgetting previous ones. A single lightweight model detects multiple diseases simultaneously. Experiments show the work achieves significantly higher accuracy and better backward transfer.

Kim et al. [108] reviewed external wearable sensors and internal implantable, injectable, and ingestible devices. The work describes how advances in low-noise electronics and flexible sensors provide rich real-time physiological data for ML models that give accurate and customized health predictions. Implantable sensors and low-power monitoring systems for precision medicine effectively manage chronic diseases.

3.6.2. QML in IoT and Wearables: Current Status

No published study demonstrates quantum advantage in IoT or wearable data processing on deployed healthcare systems. Theoretical proposals suggest using QML for multi-sensor data fusion or real-time anomaly detection. However, these proposals lack experimental validation on real IoT networks or wearable devices.

Before summarizing claimed versus demonstrated advantages, we note that performance numbers alone do not establish practical value. In healthcare, evidence also depends on dataset realism, baseline strength, and execution setting. Results from ideal simulation should be treated as feasibility tests. Results on hardware provide stronger evidence, but they still require careful reporting of shots, depth, and runtime. We use these principles when we interpret Table 6.

3.7. Summary of Demonstrated vs. Claimed QML Advantages

Table 6 consolidates claimed theoretical advantages against demonstrated results across all healthcare application domains.

4. Comparative Analysis: Classical vs. Quantum Paradigms

Section 3 demonstrated that current QML advantages are limited to controlled settings with small datasets. This section synthesizes key architectural and deployment differences between classical and quantum approaches using Healthcare 5.0 criteria. Table 7 consolidates these distinctions.

4.1. Computational Logic and Data Mapping

Classical models map data into a flat, Euclidean space. This works well for many tasks but requires large amounts of labeled data to learn complex biological links [24]. Quantum models use a much larger Hilbert space to represent data. This allows them to identify subtle correlations in genomic and proteomic datasets using fewer parameters [67]. Recent studies show that quantum kernels can sometimes separate healthy and diseased states more clearly than classical kernels in small, high-dimensional datasets [86].

4.2. Efficiency and Deployment Reality

Healthcare 5.0 emphasizes green and sustainable technology. Classical deep learning models, such as large transformers, require significant power for training and inference [109]. Quantum computing aims to reduce this energy footprint through parallel processing at the atomic level [27]. However, current quantum hardware is sensitive to noise and errors. Most real-world systems now use a hybrid model. In these systems, a classical computer handles data preparation while a quantum processor solves the most difficult parts of the problem [110].

4.3. Transition to Implementation Hurdles

The theoretical strengths of quantum models are often limited by current hardware. For example, moving medical data into a quantum state is a slow and difficult process [71]. These practical limits explain why many quantum benefits are still in the testing phase. Table 7 summarizes the key differences. These points lead directly into the technical and ethical challenges discussed in Section 5.

4.4. Resource and Runtime Reporting Template

For healthcare use, accuracy alone is not enough. Runtime, circuit resources, and sampling budgets affect latency, cost, and reproducibility. We recommend that QML studies report the items in Table 8.

5. Challenges & Limitations

Healthcare 5.0 combines AI, the Internet of Medical Things (IoMT), edge and cloud computing, and patient-centered services to support safer and more efficient care [113]. Quantum machine learning (QML) is often proposed to help with hard biomedical analytics. Examples include imaging, multimodal prediction, and optimization. However, the gap between theory and deployment is still large. Clinical systems need reliability, low latency, reproducibility, safety, privacy, and governance. Many QML studies do not yet meet these requirements [114].

We group the challenges into four layers. These are technical, algorithmic, data-centric, and socio-technical.

5.1. Technical Challenges

5.1.1. Hardware constraints: noise and decoherence in NISQ devices

Most current quantum processors are in the noisy intermediate-scale quantum (NISQ) era [115]. Qubits decohere and gates

Table 6
Claimed vs. Demonstrated QML Advantages in Healthcare Applications

Application Domain	Claimed Advantage	Demonstrated Result	Gap Analysis
Medical Imaging Classification [67, 68]	Exponential speedup via quantum feature maps	94.2% acc. (matches classical); 60% fewer params; 3x slower inference (N=500)	Speedup negated by encoding overhead and shot noise; advantage limited to parameter count on toy datasets
CT/MRI Reconstruction [69]	Quantum optimization solves inverse problems faster	15% faster convergence on 64x64 phantoms; requires fault tolerance for clinical resolution	Demonstrated only on synthetic low-res data; no path to 256x256+ clinical images on NISQ
ICU Mortality Prediction [86]	Quantum kernels capture complex interactions better	AUROC 0.847 vs. 0.842 classical (0.5% gain); 8% lower variance; 20x slower (N=15K)	Marginal accuracy improvement; computational cost prohibitive for real-time deployment
Rare Disease Cohort Matching [88]	Quantum similarity preserves structure better	12% more true positives (N=500); 50x higher computational cost	Advantage only on small cohorts; does not scale to population-level registries
Mental Health Biomarker Discovery	Quantum multi-modal fusion for fMRI+genomics	No experimental validation on patient data	Theoretical only; encoding overhead for 100K+ voxel fMRI data eliminates advantage
Clinical Decision Support	Real-time sepsis prediction via quantum pattern recognition	No published clinical validation	Theoretical only; latency requirements (<1s) incompatible with current quantum hardware
Digital Twins	Quantum simulation of drug-protein interactions in patient twins	No clinical validation or benchmarks	Theoretical only; continuous learning and real-time inference infeasible on quantum hardware
IoT/Wearable Processing	Quantum data fusion and edge anomaly detection	No experimental validation on deployed systems	Theoretical only; power budgets (mW) and edge deployment incompatible with quantum hardware

acc.: accuracy; params: parameters; AUROC: area under ROC curve; fMRI: functional MRI; NISQ: noisy intermediate-scale quantum; CT: computed tomography; MRI: magnetic resonance imaging; ICU: intensive care unit.
Key Finding: Demonstrated QML advantages are limited to small datasets (N<1000), compact embeddings (10-20 dims), and controlled settings. All real-world healthcare deployments use classical ML exclusively. No end-to-end quantum speedup has been shown on clinical cohorts (N>10K) or production systems.

Table 7
Comparative Assessment of CML and QML for Healthcare 5.0

Feature	Classical Machine Learning	Quantum Machine Learning	Key Reference
Data Geometry	Flat Euclidean space mapping.	High-dimensional Hilbert space.	ACM CSUR (2025) [111]
Parameter Use	Requires many parameters for complex tasks.	High expressivity with fewer parameters.	Nature Sci. Rep. (2024) [67]
Energy Profile	High power use for large-scale training.	Potential for low-power parallel tasks.	IEEE IoT J. (2025) [27]
Clinical Status	Fully mature and widely deployed.	Experimental and noise-sensitive.	Nature npj (2025) [24]
Best Use Case	Large datasets with clear patterns.	Small, complex, or noisy datasets.	IEEE TQE (2025) [112]

are imperfect. This limits circuit depth and reduces output stability. In healthcare, this is important because risk scores can change near clinical thresholds.

Noise modeling and mitigation are also costly. For example, simulating noise channels can be expensive [116]. Many results look strong in ideal simulation but weaken on real devices. This limits real-world use today [117].

5.1.2. Limited qubit count in NISQ devices

Today's devices offer limited usable qubits. Connectivity and noise further reduce practical capacity [118]. Healthcare data is often high dimensional. Imaging, genomics, and multimodal EHR features can exceed what we can encode and process.

Because of this, many studies compress features heavily or use small datasets. This creates uncertainty about external validity on real cohorts [119, 120].

5.1.3. Hardware–IoMT mismatch

IoMT is distributed and resource constrained. It often needs fast decisions. Quantum hardware is centralized and runs in specialized environments. In many near-term systems, quantum computation will be accessed remotely. This adds communication overhead and complicates latency guarantees [27].

Table 8
Resource and runtime template for QML reporting in healthcare studies.

Item	What to report
Data and split	Dataset name and version, sample size, patient-level split, leakage checks.
Classical baselines	Models, tuning method, compute budget, and final hyperparameters.
Encoding	Encoding type, number of features encoded, and any compression step.
Circuit size	Number of qubits, layers, and measured observables.
Circuit cost	Circuit depth and two-qubit gate count if available.
Shots	Shots per circuit, shots schedule, and estimator settings.
Training	Optimizer, learning rate, epochs, batch size, initialization, and seeds.
Backend	Hardware or simulator; if hardware, device name and date range; if simulator, noise model details.
Runtime	End-to-end runtime including encoding, transpilation, sampling, and post-processing.
Uncertainty	Mean and standard deviation across seeds; confidence intervals for metrics.

5.1.4. Circuit depth limitations

NISQ devices support only shallow circuits before noise dominates. Shallow circuits can limit model expressivity. This matters for complex healthcare signals like MRI, CT, EEG, ECG, and omics data. Deeper circuits also accumulate more gate errors. This can reduce accuracy and stability when scaling up [121].

5.2. Algorithmic Challenges

Algorithm design in QML is still maturing. Many methods are tested in simulation or on small tasks. Clinical use requires stronger evidence and better reproducibility [122].

5.2.1. Algorithmic immaturity

Many proposed QML models are not hardware-aware. Some circuits are too deep or too sensitive to noise. This causes a gap between simulator results and real-device results [111, 123].

5.2.2. Performance drop from simulation to real hardware

Simulation usually assumes ideal gates and no readout noise. Real devices have both. As a result, performance can drop after deployment to hardware. This also makes results harder to reproduce across backends [124, 125].

5.2.3. Trainability issues (barren plateaus)

Many variational models face barren plateaus. **Barren plateaus** [126] arise from the exponential concentration of gradients in deep parametrized quantum circuits. For randomly initialized circuits with L layers, the variance of cost function gradients vanishes exponentially as $\mathrm{Var}[\partial_\theta C] \in \mathcal{O}(2^{-n})$ for global cost functions, making gradient-based optimization infeasible [49]. This phenomenon originates from the formation of approximate 2-designs: deep random circuits produce output states that are nearly uniformly distributed over the Hilbert space, causing gradients to concentrate around zero. Recent mitigation strategies include (i) *correlation-aware initialization* that preserves non-zero gradient variance by avoiding uniformity [127], (ii) *layer-wise training* where shallow sub-circuits are optimized sequentially before adding depth [128], and (iii) *local cost functions* that measure observables on limited qubit subsets [49]. For healthcare VQCs, these strategies reduce trainability issues but do not eliminate them entirely; shallow circuits ($L \leq 10$) remain the practical regime for NISQ applications.

Gradients can become very small as circuits grow. Training becomes slow and unstable [129, 41]. In healthcare, this is risky because models must be stable across runs and datasets.

5.2.4. Lack of clinical integration workflows

Even accurate models must fit clinical workflows. This includes data ingestion, logging, audit trails, calibration checks, monitoring, and safe fallback. Classical medical AI has mature pathways in some areas, such as PACS integration in imaging. QML pipelines rarely describe these deployment details. This limits clinical translation [130, 131].

5.3. Data-Centric Challenges

5.3.1. Data encoding cost

Data encoding is a major bottleneck for QML. Healthcare data must be converted into quantum states. Many encoding methods require deep circuits, repeated preparation, and normalization. Encoding can dominate runtime and amplify noise [132, 133]. This is a key reason why end-to-end quantum advantage is hard to show in practice.

5.3.2. High-dimensional healthcare data

Healthcare data is multimodal and high dimensional. It includes images, time series, notes, omics data, and IoMT streams. Limited qubits force compression or feature selection. This can remove clinically relevant information and reduce generalization [134].

5.3.3. Noise during state preparation

Noise can corrupt information during state preparation. Errors at encoding can propagate through the circuit. Current devices do not have full error correction. This can distort predictions and uncertainty estimates [135].

5.3.4. Loss of clinically relevant features

Compression and encoding choices can remove rare but important signals. This can hurt performance for small subgroups and edge cases. It can also increase bias risk when data is imbalanced [136].

Table 9
State-of-the-Art Challenges of Quantum Machine Learning in Healthcare 5.0

Challenge Layer	Core Challenges	Root Causes	Main Technical Impact	Healthcare 5.0 Impact
Quantum Hardware [17, 137]	Noise, decoherence, limited qubit availability, lack of fault tolerance	NISQ-era hardware; immature error mitigation; limited access to scalable quantum devices	High error rates; shallow circuits; unreliable outputs; restricted model complexity	Reduced diagnostic reliability; inability to process high-dimensional EHR, imaging, and genomic data
Algorithmic [49, 138]	Barren plateaus; dominance of linear QML models; limited expressibility under noise	Noise-aware ansätze; vanishing gradients; restrictive NISQ training assumptions	Training instability; poor convergence; absence of quantum advantage	Weak clinical decision support; limited justification in safety-critical environments
Data Encoding [139, 140, 141]	High qubit demand; costly state preparation; scalability failure	Amplitude and angle encoding overhead; normalization constraints; circuit depth limits	Exponential resource growth; amplified noise sensitivity; degraded class separability	Latency incompatible with real-time clinical workflows; infeasible large-scale deployment
Ethical & Societal [142, 19, 138]	Privacy risks; explainability gaps; bias and equity concerns	Probabilistic quantum outputs; lack of QML-specific XAI; high infrastructure cost	Limited interpretability; data security vulnerabilities; unequal access	Loss of patient trust; accountability gaps; widening global health disparities

5.4. Socio-Technical Challenges

Ethics Healthcare deployment requires transparency and accountability. QML outputs are probabilistic and models can be hard to interpret. Bias can also occur if data is imbalanced. Access is another concern. Quantum infrastructure is costly, and this can widen gaps between institutions [143].

Privacy Healthcare data is sensitive. Many QML workflows rely on remote quantum execution. This can require sending data or embeddings outside the hospital. It increases governance and security risk. Faster quantum progress also creates uncertainty about long-term security assumptions [144].

Quantum decryption threats Future quantum capabilities may break common public-key cryptography. This creates a long-term risk for archived medical data. Health records stored today could be decrypted later. This supports the need for post-quantum security planning [145].

Regulatory concerns Regulatory pathways for QML systems are still unclear. Existing rules such as GDPR and HIPAA do not directly specify how to audit sampling variability and error mitigation. Clinical adoption needs clearer validation practices and governance for quantum-enabled pipelines [146].

All these issues interact [147]. Hardware limits constrain circuits. Circuit limits shape algorithms. Algorithms drive encoding choices. These choices affect privacy, auditability, and clinical trust.

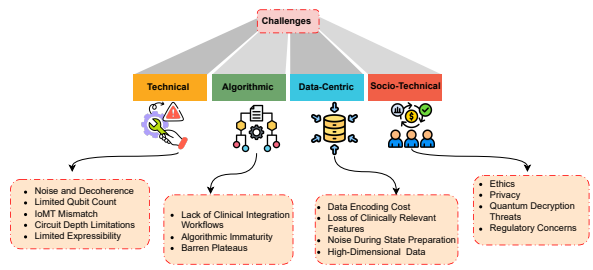

Figure 5: Overview of key challenges in QML for Healthcare 5.0 across technical, algorithmic, data-centric, and socio-technical layers.

The overview of key challenges in QML for Healthcare 5.0 across technical, algorithmic, data-centric, and socio-technical layers are Fig. 5

6. Future Research Directions

Future work should move beyond proof-of-concept studies and small datasets [114, 24]. It should also match the deployment reality of NISQ hardware and hybrid workflows [27]. The goal is to build methods that are scalable, reproducible, and clinically useful.

6.1. Near-term directions: hybrid QML and better benchmarks

In the near term, hybrid pipelines are the most realistic path. A classical model can handle data cleaning and feature extraction. A small quantum module can handle a narrow

decision or optimization step. This matches what we see in imaging and prediction pipelines.

The community also needs shared benchmarks. Benchmarks should include strong classical baselines. They should report accuracy, calibration, and robustness. They should also report encoding cost, runtime, and sampling budgets. Benchmarks should include imaging, EHR prediction, multi-omics, and IoMT streams. This will help the field avoid claims based on toy settings.

6.2. Mid-term directions: encoding, scaling, and workflow integration

Mid-term progress depends on encoding-aware design. Encoding should use the structure of healthcare data. It should keep qubit counts small and circuits shallow. It should preserve clinically meaningful signals.

Workflow integration is also required. QML needs secure service interfaces that hospitals can use. It needs logging for both quantum and classical parts. It needs monitoring for drift and calibration. It needs clear rollback and fallback rules. Without this, QML will remain a lab demo rather than a clinical tool.

6.3. Long-term directions: fault-tolerant QML ecosystems

In the long term, fault-tolerant hardware will change what is possible. It could support deeper circuits and larger logical qubit counts. This could enable stronger quantum-native learning and generative modeling. It could also support richer digital twins that simulate disease progression and treatment response.

Even then, clinical acceptance will require interpretability, safety, and equity. The field will also need collaboration between quantum engineers, clinicians, and regulators.

6.4. Open problems: benchmarks and quantum federated learning

Two open problems matter across all domains.

First, we still lack standardized QML benchmarks in healthcare. Many studies use private data or custom splits. Results are hard to compare. Community benchmark suites and clear protocols would improve reproducibility and fairness evaluation.

Second, quantum federated learning (QFL) is promising but early. Healthcare data is distributed across hospitals and devices. Privacy rules often prevent central pooling. QFL aims to train models without sharing raw data. Open questions include aggregation design, privacy guarantees under quantum adversaries, and integration with post-quantum cryptography and classical privacy tools. These topics are important for Healthcare 5.0.

Figure 6 summarizes the main future directions.

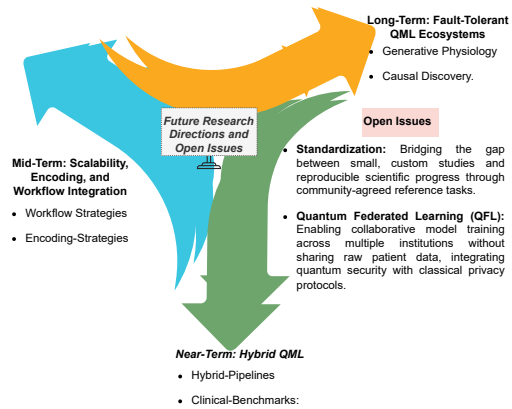

Figure 6: Future research directions in QML for Healthcare 5.0.

7. Benchmarking and Reporting Checklist

Many QML studies in healthcare are hard to compare because they use different datasets, different splits, and different baselines. Some papers also omit key engineering details, such as shots, circuit depth, and runtime. This section provides a short checklist for authors and reviewers. The goal is to support fair comparison and reproducibility.

We recommend reporting items in five groups. These are task and data, baselines, quantum model, training, and execution setting. When an item is not applicable, the paper should state this clearly. Table 10 lists the minimum items.

8. Conclusion

Healthcare 5.0 aims to deliver care that is more personal, safe, and efficient by combining AI, connected sensing, edge and cloud computing, and digital twins. This survey reviewed how quantum machine learning can support this goal, and where it still falls short. We summarized the main QML approaches used in healthcare, including quantum kernels, variational quantum circuits, and hybrid quantum classical pipelines. Across imaging, EHR-based prediction, remote monitoring, clinical decision support, neuroinformatics, and patient digital twins, the most realistic near-term pattern is hybrid use. In this pattern, classical models handle feature extraction and data cleaning, while a small quantum component is used as a decision layer or as an optimizer for a hard subproblem.

The comparison between classical and quantum paradigms shows a clear tradeoff. Classical machine learning is mature, fast to deploy, and supported by strong tools and clinical workflows. Quantum models offer new feature spaces and may help with some high dimensional and noisy problems, especially when labels are limited. However, current benefits are often reduced by data encoding cost, repeated circuit execution, and hardware noise. Many published studies also use small datasets and weak baselines, which makes results hard to trust and hard to reproduce. For healthcare, this matters because models must be reliable, fair, and well calibrated near clinical decision thresholds.

Based on this evidence, QML is not ready to replace classical systems in real hospitals. Its value today is as a focused add-on inside a larger classical pipeline. Future progress depends on better encoding methods, noise-aware

Table 10
Minimum benchmarking and reporting checklist for QML in healthcare.

Group	Minimum items to report
Task and data	Task definition; dataset name and version; sample size; split strategy; patient-level split when relevant; preprocessing; metrics with confidence intervals; calibration metric when used for risk scores.
Baselines	At least one linear and one strong non-linear baseline; tuned hyperparameters; compute budget; fairness statement for comparison.
Quantum model	Method class (kernel, VQC, optimization, generative, hybrid); encoding method; qubits; depth or layers; measurement scheme; output mapping; stated constraints on scaling.
Training	Optimizer; learning rate; epochs; batch size; initialization; number of seeds; mean and standard deviation; early stopping rule if used.
Execution setting	Hardware or simulator; noise model if simulated; device details and date range if hardware; transpilation settings; shots; total number of circuit evaluations; end-to-end runtime.

training, and careful reporting of runtime and resource use. It also depends on stronger evaluation practice, including shared benchmarks, external validation across sites, and clear comparison against well tuned classical models. If the community follows these steps, QML can move from small demonstrations toward tools that can improve robustness, calibration, and decision quality in selected Healthcare 5.0 tasks.

References

[1] H. Ossebaard, ihealth: supporting health by technology, Phd thesis - research ut, graduation ut, University of Twente (Jun. 2012). doi:10.3990/1.9789036533751.

[2] V. Puri, A. Kataria, V. Sharma, Artificial intelligence-powered decentralized framework for internet of things in healthcare 4.0, Transactions on Emerging Telecommunications Technologies 35 (4) (2024) e4245. arXiv:https://onlinelibrary.wiley.com/doi/pdf/10.1002/ett.4245, doi:https://doi.org/10.1002/ett.4245.
URL https://onlinelibrary.wiley.com/doi/abs/10.1002/ett.4245

[3] A. Ibrahim Adamu, P. Kumar Donta, D. Mohd Ali, S. Sarang, G. M. Stojanović, S. Seroja Sarnin, A systematic literature review of advanced machine learning techniques in wireless body area networks: Application, challenges, and future directions, IEEE Access 13 (2025) 194729–194778. doi:10.1109/ACCESS.2025.3631230.

[4] Ö. Sabuncu, B. Bilgehan, Revolutionizing healthcare 5.0: Blockchain-driven optimization of drone-to-everything communication using 5g network for enhanced medical services, Technology in Society 77 (2024) 102552. doi:10.1016/j.techsoc.2024.102552.

[5] A. Younesi, E. Oustad, M. Ansari, T. Fahringer, R. Buyya, Healthcare 5.0: An industry 5.0 perspective for next-generation medical systems with synergistic integration of iot, ai, and 6g, Internet of Things (2025) 101815doi:10.1016/j.iot.2025.101815.

[6] D. Saraswat, P. Bhattacharya, A. Verma, V. K. Prasad, S. Tanwar, G. Sharma, P. N. Bokoro, R. Sharma, Explainable ai for healthcare 5.0: Opportunities and challenges, IEEE Access 10 (2022) 84486–84517. doi:10.1109/ACCESS.2022.3197671.

[7] K. R. Blenman, R. Hull, C. Maimela, T. Molefi, R. Khanyile, Z. Dlamini, Society 5.0 healthcare: Ethics, legal rights, human rights, safety and security, in: Society 5.0 and Next Generation Healthcare: Patient-Focused and Technology-Assisted Precision Therapies, Springer, 2023, pp. 267–291. doi:10.1007/978-3-031-36461-7_12.

[8] E. J. Topol, High-performance medicine: the convergence of human and artificial intelligence, Nature medicine 25 (1) (2019) 44–56. doi:10.1038/s41591-018-0300-7.

[9] P. Rajpurkar, E. Chen, O. Banerjee, E. J. Topol, Ai in health and medicine, Nature medicine 28 (1) (2022) 31–38. doi:10.1038/s41591-021-01614-0.

[10] A. Holzinger, G. Langs, H. Denk, K. Zatloukal, H. Muller, Causability and explainability of artificial intelligence in medicine, WIREs Data Mining and Knowledge Discovery 9 (4) (2019) e1312. arXiv:https://wires.onlinelibrary.wiley.com/doi/pdf/10.1002/widm.1312, doi:10.1002/widm.1312.
URL https://wires.onlinelibrary.wiley.com/doi/abs/10.1002/widm.1312

[11] M. Sahu, P. Maurya, S. Krishnaveni, Healthcare 5.0: enabling technologies and potential use cases, in: Industry 5.0: Key Technologies and Drivers, Springer, 2025, pp. 303–327.

[12] M. Z. Riaz, B. K. Behera, S. Mumtaz, S. Al-Kuwari, A. Farouk, Quantum machine learning for energy-efficient 5g-enabled iomt healthcare systems: Enhancing data security and processing, IEEE Internet of Things Journal 12 (22) (2025) 46119–46128. doi:10.1109/JIOT.2025.3586090.

[13] M. Al-Hawawreh, M. S. Hossain, A human-centered quantum machine learning framework for attack detection in iot-based healthcare industry 5.0, IEEE Internet of Things Journal 12 (22) (2025) 46065–46074. doi:10.1109/JIOT.2025.3565687.

[14] S. Chakraborty, A study on quantum neural networks in healthcare 5.0 (2024). arXiv:2412.06818.
URL https://arxiv.org/abs/2412.06818

[15] A. S. Ashour, D. Koundal, Quantum computing in healthcare 5.0, in: Quantum Computing for Healthcare Data, Elsevier, 2025, pp. 43–62.

[16] V. K. Bukkarayasamudram, P. C. S. Reddy, K. Arun Kumar, R. Jagadish, S. Sharma, M. L. Prasad, Y. Sucharitha, I. A. Tayubi, G. K. Thakur, Quantum computing revolution in healthcare: a systematic review of applications, issues and future directions, Artificial Intelligence Review 58 (12) (2025) 1–35.

[17] R. Ur Rasool, H. F. Ahmad, W. Rafique, A. Qayyum, J. Qadir, Z. Anwar, Quantum computing for healthcare: A review, Future Internet 15 (3) (2023) 94.

[18] A. Marengo, V. Santamato, Quantum algorithms and complexity in healthcare applications: a systematic review with machine learning-optimized analysis, Frontiers in Computer Science 7 (2025) 1584114.

[19] R. S. Gupta, C. E. Wood, T. Engstrom, J. D. Pole, S. Shrapnel, A systematic review of quantum machine learning for digital health, npj Digital Medicine 8 (1) (2025) 237.

[20] J. Biamonte, P. Wittek, N. Pancotti, P. Rebentrost, N. Wiebe, S. Lloyd, Quantum machine learning, Nature 549 (7671) (2017) 195–202. doi:10.1038/nature23474.

[21] M. Schuld, I. Sinayskiy, F. Petruccione, An introduction to quantum machine learning, Contemporary Physics 56 (2) (2015) 172–185. doi:10.1080/00107514.2014.964942.

[22] D. Peral-García, J. Cruz-Benito, F. J. García-Peñalvo, Systematic literature review: Quantum machine learning and its applications, Computer Science Review 51 (2024) 100619. doi:10.1016/j.cosrev.2024.100619.

[23] J. C. L. Chow, Quantum computing and machine learning in medical decision-making: A comprehensive review, Algorithms 18 (3) (2025) 156. doi:10.3390/a18030156.

[24] R. S. Gupta, C. E. Wood, T. Engstrom, J. D. Pole, S. Shrapnel, A systematic review of quantum machine learning for digital health, npj Digital Medicine 8 (1) (2025) 237.

[25] S. Chakraborty, A study on quantum neural networks in healthcare 5.0, arXiv preprint arXiv:2412.06818 (2024).

[26] M. Al-Hawawreh, M. S. Hossain, A human-centered quantum machine learning framework for attack detection in iot-based healthcare industry 5.0, IEEE Internet of Things Journal (2025).

[27] M. Z. Riaz, B. K. Behera, S. Mumtaz, S. Al-Kuwari, A. Farouk, Quantum machine learning for energy-efficient 5g-enabled iomt healthcare systems: Enhancing data security and processing, IEEE Internet of Things Journal (2025).

[28] R. Tharsanee, B. Vishnuupriya, C. A. D. Durai, The future of quantum computing in healthcare, in: Quantum Computing for Healthcare Data, Elsevier, 2025, pp. 229–251.

[29] A. K. Selvaraj, K. S. Kannan, S. R. Krishnamoorthy, D. K. Anandhan, S. B. Prathiba, D. K. Sah, P. K. Donta, Quarta-6g: Unified post-quantum security and quantum learning for uavs in 6g iot, IEEE Internet of Things Journal (2025) 1–1doi:10.1109/JIOT.2025.3631644.

[30] E. Zeydan, C. De Alwis, R. Khan, Y. Turk, A. Aydeger, T. R. Gadekallu, M. Liyanage, Quantum technologies for beyond 5g and 6g networks: Applications, opportunities, and challenges, arXiv preprint arXiv:2504.17133 (2025).

[31] R. Kumar, M. Swarnkar, Quids: A quantum support vector machine-based intrusion detection system for iot networks, Journal of Network and Computer Applications 234 (2025) 104072.

[32] L. Chen, T. Li, Y. Chen, X. Chen, M. Wozniak, N. Xiong, W. Liang, Design and analysis of quantum machine learning: a survey, Connection Science 36 (1) (2024) 2312121.

[33] T. M. Forcer, A. J. Hey, D. Ross, P. G. Smith, Superposition, entanglement and quantum computation., Quantum Inf. Comput. 2 (2) (2002) 97–116.

[34] A. W. Harrow, A. Hassidim, S. Lloyd, Quantum algorithm for linear systems of equations, Physical review letters 103 (15) (2009) 150502.

[35] P. Lamichhane, D. B. Rawat, Quantum machine learning: Recent advances, challenges and perspectives, IEEE Access (2025).

[36] M. Akrom, Quantum support vector machine for classification task: A review, Journal of Multiscale Materials Informatics 1 (2) (2024) 1–8.

[37] M. Cerezo, G. Verdon, H.-Y. Huang, L. Cincio, P. J. Coles, Challenges and opportunities in quantum machine learning, Nature computational science 2 (9) (2022) 567–576.

[38] H. Liu, S. Zhang, M. Li, W. Liang, V. K. Arthur Sandor, A real-time privacy-preserving scheme based on grouping queries for continuous location-based services, Concurrency and Computation: Practice and Experience 35 (19) (2023) e7404.

[39] A. Callison, N. Chancellor, Hybrid quantum-classical algorithms in the noisy intermediate-scale quantum era and beyond, Physical Review A 106 (1) (2022) 010101.

[40] L. Fan, Z. Han, Hybrid quantum-classical computing for future network optimization, IEEE Network 36 (5) (2022) 72–76.

[41] C.-Y. Liu, E.-J. Kuo, C.-H. Abraham Lin, J. Gemsun Young, Y.-J. Chang, M.-H. Hsieh, H.-S. Goan, Quantum-train: Rethinking hybrid quantum-classical machine learning in the model compression perspective, Quantum Machine Intelligence 7 (2) (2025) 80.

[42] V. Havlíček, A. D. Córcoles, K. Temme, A. W. Harrow, A. Kandala, J. M. Chow, J. M. Gambetta, Supervised learning with quantum-enhanced feature spaces, Nature 567 (7747) (2019) 209–212.

[43] V. Giovannetti, S. Lloyd, L. Maccone, Quantum random access memory, Physical review letters 100 (16) (2008) 160501.

[44] E. Tang, A quantum-inspired classical algorithm for recommendation systems, in: Proceedings of the 51st annual ACM SIGACT symposium on theory of computing, 2019, pp. 217–228.

[45] H.-Y. Huang, M. Broughton, M. Mohseni, R. Babbush, S. Boixo, H. Neven, J. R. McClean, Power of data in quantum machine learning, Nature communications 12 (1) (2021) 2631.

[46] M. Schuld, R. Sweke, J. J. Meyer, Effect of data encoding on the expressive power of variational quantum-machine-learning models, Physical Review A 103 (3) (2021) 032430.

[47] A. S. Holevo, Bounds for the quantity of information transmitted by a quantum communication channel, Problemy Peredachi Informatsii 9 (3) (1973) 3–11.

[48] W. K. Wootters, W. H. Zurek, A single quantum cannot be cloned, Nature 299 (5886) (1982) 802–803.

[49] M. Cerezo, A. Sone, T. Volkoff, L. Cincio, P. J. Coles, Cost function dependent barren plateaus in shallow parametrized quantum circuits, Nature communications 12 (1) (2021) 1791.

[50] J. Irvin, P. Rajpurkar, M. Ko, Y. Yu, S. Ciurea-Ilcus, C. Chute, H. Marklund, B. Haghgoo, R. Ball, K. Shpanskaya, et al., Chexpert: A large chest radiograph dataset with uncertainty labels and expert comparison, in: Proceedings of the AAAI conference on artificial intelligence, Vol. 33, 2019, pp. 590–597.

[51] X. Mei, Z. Liu, P. M. Robson, B. Marinelli, M. Huang, A. Doshi, A. Jacobi, C. Cao, K. E. Link, T. Yang, et al., Radimagenet: an open radiologic deep learning research dataset for effective transfer learning, Radiology: Artificial Intelligence 4 (5) (2022) e210315.

[52] S. A. Lee, A. Wu, J. N. Chiang, Clinical modernbert: An efficient and long context encoder for biomedical text (2025). arXiv:2504.03964. URL https://arxiv.org/abs/2504.03964

[53] A. E. Johnson, T. J. Pollard, L. Shen, L.-w. H. Lehman, M. Feng, M. Ghassemi, B. Moody, P. Szolovits, L. Anthony Celi, R. G. Mark, Mimic-iii, a freely accessible critical care database, Scientific data 3 (1) (2016) 1–9.

[54] T. J. Pollard, A. E. Johnson, J. D. Raffa, L. A. Celi, R. G. Mark, O. Badawi, The eicu collaborative research database, a freely available multi-center database for critical care research, Scientific data 5 (1) (2018) 1–13.

[55] F. Isensee, P. F. Jaeger, S. A. Kohl, J. Petersen, K. H. Maier-Hein, nnu-net: a self-configuring method for deep learning-based biomedical image segmentation, Nature methods 18 (2) (2021) 203–211.

[56] H. Harutyunyan, H. Khachatrian, D. C. Kale, G. Ver Steeg, A. Galstyan, Multitask learning and benchmarking with clinical time series data, Scientific data 6 (1) (2019) 96.

[57] L. Rasmy, Y. Xiang, Z. Xie, C. Tao, D. Zhi, Med-bert: pre-trained contextualized embeddings on large-scale structured electronic health records for disease prediction, NPJ digital medicine 4 (1) (2021) 86.

[58] Y. Li, S. Rao, J. R. A. Solares, A. Hassaine, R. Ramakrishnan, D. Canoy, Y. Zhu, K. Rahimi, G. Salimi-Khorshidi, Behrt: transformer for electronic health records, Scientific reports 10 (1) (2020) 7155.

[59] O. Esteban, C. J. Markiewicz, R. W. Blair, C. A. Moodie, A. I. Isik, A. Erramuzpe, J. D. Kent, M. Goncalves, E. DuPre, M. Snyder, et al., fmriprep: a robust preprocessing pipeline for functional mri, Nature methods 16 (1) (2019) 111–116.

[60] B. Fischl, Freesurfer, Neuroimage 62 (2) (2012) 774–781.

[61] J. N. Kather, J. Krisam, P. Charoentong, T. Luedde, E. Herpel, C.-A. Weis, T. Gaiser, A. Marx, N. A. Valous, D. Ferber, et al., Predicting survival from colorectal cancer histology slides using deep learning: A retrospective multicenter study, PLoS medicine 16 (1) (2019) e1002730.

[62] G. Campanella, M. G. Hanna, L. Geneslaw, A. Miraflor, V. Werneck Krauss Silva, K. J. Busam, E. Brogi, V. E. Reuter, D. S. Klimstra, T. J. Fuchs, Clinical-grade computational pathology using weakly supervised deep learning on whole slide images, Nature medicine 25 (8) (2019) 1301–1309.

[63] F. Shamshad, S. Khan, S. W. Zamir, M. H. Khan, M. Hayat, F. S. Khan, H. Fu, Transformers in medical imaging: A survey, Medical Image Analysis 88 (2023) 102802. doi:https://doi.org/10.1016/j.media.2023.102802.
URL https://www.sciencedirect.com/science/article/pii/S1361841523000634

[64] K. Dembrower, E. Wåhlin, Y. Liu, M. Salim, K. Smith, P. Lindholm, M. Eklund, F. Strand, Effect of artificial intelligence-based triaging of breast cancer screening mammograms on cancer detection and radiologist workload: a retrospective simulation study, The Lancet Digital Health 2 (9) (2020) e468–e474.

[65] S. Takahashi, Y. Sakaguchi, N. Kouno, K. Takasawa, K. Ishizu, Y. Akagi, R. Aoyama, N. Teraya, A. Bolatkan, N. Shinkai, et al., Comparison of vision transformers and convolutional neural networks in medical image analysis: A systematic review, Journal of Medical Systems 48 (1) (2024) 84.

[66] H. Zhang, Y. Qie, Applying deep learning to medical imaging: A review, Applied Sciences 13 (18) (2023). doi:10.3390/app131810521.
URL https://www.mdpi.com/2076-3417/13/18/10521

[67] Q. Xiang, D. Li, Z. Hu, Y. Yuan, Y. Sun, Y. Zhu, Y. Fu, Y. Jiang, X. Hua, Quantum classical hybrid convolutional neural networks for breast cancer diagnosis, Scientific Reports 14 (1) (2024) 24699.

[68] F. Fan, Y. Shi, T. Guggemos, X. X. Zhu, Hybrid quantum-classical convolutional neural network model for image classification, IEEE Transactions on Neural Networks and Learning Systems 35 (12) (2024) 18145–18159. doi:10.1109/TNNLS.2023.3312170.

[69] K. Dremel, D. Prjamkov, M. Firsching, M. Weule, T. Lang, A. Papadaki, S. Kasperl, M. Blaimer, T. O. J. Fuchs, Utilizing quantum annealing in computed tomography image reconstruction, IEEE Transactions on Quantum Engineering 6 (2025) 1–10. doi:10.1109/TQE.2025.3549305.

[70] P. Nandal, S. Pahal, G. M. Upadhyay, Image denoising using quantum deep convolutional generative adversarial network for medical images, International Journal of Computational Intelligence Systems 18 (1) (2025) 190.

[71] M. Rath, H. Date, Quantum data encoding: A comparative analysis of classical-to-quantum mapping techniques and their impact on machine learning accuracy, EPJ Quantum Technology 11 (1) (2024) 72.

[72] P. de Alcântara, G. Audi, L. Morais, Lie groups for quantum complexity and barren plateau theory (2025).

[73] Z. Krunic, F. F. Flöther, G. Seegan, N. D. Earnest-Noble, O. Shehab, Quantum kernels for real-world predictions based on electronic health records, IEEE Transactions on Quantum Engineering 3 (2022) 1–11. doi:10.1109/TQE.2022.3176806.

[74] C.-S. Chen, W. A. Hou, H.-W. Hu, Z.-S. Cai, Quantum generative models for image generation: Insights from mnist and medmnist (2025). arXiv:2504.00034.
URL https://arxiv.org/abs/2504.00034

[75] L. Wei, H. Liu, J. Xu, L. Shi, Z. Shan, B. Zhao, Y. Gao, Quantum machine learning in medical image analysis: A survey, Neurocomputing 525 (2023) 42–53. doi:https://doi.org/10.1016/j.neucom.2023.01.049.
URL https://www.sciencedirect.com/science/article/pii/S0925231223000589

[76] P. K. Shukla, S. Stalin, S. Joshi, P. K. Shukla, P. K. Pareek, Optimization assisted bidirectional gated recurrent unit for healthcare monitoring system in big-data, Applied Soft Computing 138 (2023) 110178.

[77] Z. Wang, S. Chen, T. Liu, B. Yao, Multi-branching temporal convolutional network with tensor data completion for diabetic retinopathy prediction, IEEE Journal of Biomedical and Health Informatics 28 (3) (2024) 1704–1715. doi:10.1109/JBHI.2024.3351949.

[78] C. A. Siebra, M. Kurpicz-Briki, K. Wac, Transformers in health: a systematic review on architectures for longitudinal data analysis, Artificial Intelligence Review 57 (2) (2024) 32.

[79] J. Lee, W. Yoon, S. Kim, D. Kim, S. Kim, C. H. So, J. Kang, Biobert: a pre-trained biomedical language representation model for biomedical text mining, Bioinformatics 36 (4) (2019) 1234–1240. arXiv:https://academic.oup.com/bioinformatics/article-pdf/36/4/1234/48983216/bioinformatics_36_4_1234.pdf, doi:10.1093/bioinformatics/btz682.
URL https://doi.org/10.1093/bioinformatics/btz682

[80] S. Wang, J. Bian, X. Huang, H. Zhou, S. Zhu, Publabeler: Enhancing automatic classification of publications in uniprotkb using protein textual description and pubmedbert, IEEE Journal of Biomedical and Health Informatics 29 (5) (2025) 3782–3791. doi:10.1109/JBHI.2024.3520579.

[81] H. Cao, Y. Wang, J. Chen, D. Jiang, X. Zhang, Q. Tian, M. Wang, Swin-unet: Unet-like pure transformer for medical image segmentation, in: European conference on computer vision, Springer, 2022, pp. 205–218.

[82] V. K. Prasad, A. Verma, P. Bhattacharya, S. Shah, S. Chowdhury, M. Bhavsar, S. Aslam, N. Ashraf, Revolutionizing healthcare: a comparative insight into deep learning's role in medical imaging, Scientific Reports 14 (1) (2024) 30273.

[83] A. Al-Dailami, H. Kuang, J. Wang, Multimodal representation learning based on personalized graph-based fusion for mortality prediction using electronic medical records, Big Data Mining and Analytics 8 (4) (2025) 933–950. doi:10.26599/BDMA.2024.9020099.

[84] S. Chi, Y. Tian, X. Li, F. Wang, Y. Wang, T. Zhou, P. Zhang, J. Chen, J. Li, A self-supervised graph neural network to identify temporal phenotypes of end-stage renal disease using longitudinal electronic health records, Journal of Healthcare Informatics Research (2025) 1–25.

[85] R. Lin, Ai-driven personalized healthcare: Leveraging multimodal data for precision medicine, in: 2024 IEEE/WIC International Conference on Web Intelligence and Intelligent Agent Technology (WI-IAT), IEEE, 2024, pp. 693–697.

[86] J. Schnabel, M. Roth, Quantum kernel methods under scrutiny: a benchmarking study, Quantum Machine Intelligence 7 (1) (2025) 58.

[87] A. Abbas, D. Sutter, C. Zoufal, A. Lucchi, A. Figalli, S. Woerner, The power of quantum neural networks, Nature Computational Science 1 (6) (2021) 403–409.

[88] M. Schuld, N. Killoran, Quantum machine learning in feature hilbert spaces, Physical review letters 122 (4) (2019) 040504.

[89] K. Ouanes, F. Nesren, Effectiveness of artificial intelligence (ai) in clinical decision support systems and care delivery, Journal of medical systems (2024) 48.

[90] V. Santamato, C. Tricase, N. Faccilongo, M. Iacoviello, A. Marengo, Exploring the impact of artificial intelligence on healthcare management: a combined systematic review and machine-learning approach, Applied Sciences 14 (2024) 10144.

[91] Z. Zhou, B. Chen, Z.-J. Mei, W. Chen, W. Cao, E.-X. Xu, J. Wang, L. Ye, H.-W. Cheng, Machine learning-based scoring model for predicting mortality in icu-admitted ischemic stroke patients with moderate to severe consciousness disorders., Frontiers in Neurology 16 (2025) 1534961.

[92] P. Gallego, A. Eva, P. Jaime, J. Nuria, A. Cristina, T. Natalia, V. Helena, Breast radiotherapy planning: A decision-making framework using deep learning, Medical Physics 52 (2025) 1798–1809.

[93] L. Wang, X. Guo, H. Shi, Y. Ma, H. Bao, L. Jiang, L. Zhao, Z. Feng, T. Zhu, L. Lu, Crisp: A causal relationships-guided deep learning framework for advanced icu mortality prediction, BMC Medical Informatics and Decision Making 25 (2025) 165.

[94] S. Bozyel, E. Şimşek, D. Koçyiğit, A. Güler, Y. Korkmaz, M. Şeker, M. Ertürk, N. Keser, Artificial intelligence-based clinical decision support systems in cardiovascular diseases, Anatolian journal of cardiology 28 (2024) 74.

[95] M. Abrams, V. H. John Darrell, Bridging the gap: How neuroinformatics is preparing the next generation of neuroscience researchers,

Neuroinformatics 22 (2024) 619–622.

[96] T. M. Zakaria, A. Z. Langi, N. M. Sophian, I. Anshori, Artificial intelligence (ai) in neurofeedback therapy using electroencephalography (eeg), heart rate variability (hrv), galvanic skin response (gsr)., IEEE Access 13 (2025) 133078 – 133112.

[97] S. Vieira, L. Baecker, W. Pinaya, R. Garcia-Dias, C. Scarpazza, V. Calhoun, A. Mechelli, Neurofind: using deep learning to make individualised inferences in brain-based disorders, Translational psychiatry 15 (2025) 69.

[98] M. D. Nath, M. K. U. Ahamed, O. Ahmed, T. Ahmed, S. Roy, M. N. Uddin, Smart web interface for student mental health prediction using machine learning with blockchain technology, Neuroscience Informatics 5 (2025) 100236.

[99] F. A. Zeiler, Y. Iturria-Medina, E. P. Thelin, A. Gomez, J. J. Shankar, J. H. Ko, C. R. Figley, G. E. Wright, C. M. Anderson, Integrative neuroinformatics for precision prognostication and personalized therapeutics in moderate and severe traumatic brain injury, Frontiers in neurology 12 (2021) 729184.

[100] M. D. Xames, G. T. Taylan, A systematic literature review of digital twin research for healthcare systems: Research trends, gaps, and realization challenges, IEEE access 12 (2024) 4099–4126.

[101] R. Brahmi, B. Noureddine, E. Ridha, Elaboration of innovative digital twin models for healthcare monitoring with 6g functionalities, IEEE Access (109608 - 109624) 12.

[102] P. M. Thangaraj, S. H. Benson, E. K. Oikonomou, F. W. Asselbergs, R. Khera, Cardiovascular care with digital twin technology in the era of generative artificial intelligence, European Heart Journal 45 (2024) 4808–4821.

[103] S. Ullah, S. Khan, D. Vanecek, I. U. Rehman, Machine learning and digital-twins-based internet of robotic things for remote patient monitoring, IEEE Access (2025).

[104] A. D. Patel, R. H. Jhaveri, A. D. Patel, K. A. Shah, J. Shah, Enhanced aiot multi-modal fusion for human activity recognition in ambient assisted living environment, Software: Practice and Experience 55 (2025) 731–747.

[105] B. Zhou, G. Yang, Z. Shi, S. Ma, Natural language processing for smart healthcare, IEEE Reviews in Biomedical Engineering 17 (2022) 4–18.

[106] H. Gao, Y. Wan, H. Xu, L. Chen, J. Xiao, Q. Ran, Swinbtc: Transfer learning to brain tumor classification for healthcare electronics using augmented mr images, IEEE Transactions on Consumer Electronics (2025).

[107] C.-H. Li, J. Niraj K., Doctor: A multi-disease detection continual learning framework based on wearable medical sensors, ACM Transactions on Embedded Computing Systems 23 (2024) 1–33.

[108] A. Koruprolu, T. Hack, O. Ghadami, A. Jain, D. A. Hall, From wearables to implantables: Harnessing sensor technologies for continuous health monitoring, IEEE Transactions on Biomedical Circuits and Systems (2025).

[109] G. Yenduri, M. Ramalingam, G. C. Selvi, Y. Supriya, G. Srivastava, P. K. R. Maddikunta, G. D. Raj, R. H. Jhaveri, B. Prabadevi, W. Wang, et al., Gpt (generative pre-trained transformer)—a comprehensive review on enabling technologies, potential applications, emerging challenges, and future directions, IEEE access 12 (2024) 54608–54649.

[110] F. Fan, Y. Shi, T. Guggemos, X. X. Zhu, Hybrid quantum-classical convolutional neural network model for image classification, IEEE Transactions on Neural Networks and Learning Systems 35 (12) (2023) 18145–18159. doi:10.1109/TNNLS.2023.3312170.

[111] F. Rodríguez-Díaz, D. Gutiérrez-Avilés, A. Troncoso, F. Martínez-Álvarez, A survey of quantum machine learning: Foundations, algorithms, frameworks, data and applications, ACM Computing Surveys 58 (4) (2025) 1–35. doi:10.1145/3708514.

[112] K. Dremel, D. Prjamkov, M. Firsching, M. Weule, T. Lang, A. Papadaki, S. Kasperl, M. Blaimer, T. O. J. Fuchs, Utilizing quantum annealing in computed tomography image reconstruction, IEEE Transactions on Quantum Engineering 6 (2025) 1–10. doi:10.1109/TQE.2025.3549305.

[113] J.-C. Gamazo-Real, R. T. Fernández, A. M. Armas, Comparison of edge computing methods in internet of things architectures for efficient estimation of indoor environmental parameters with machine learning, Engineering Applications of Artificial Intelligence 126 (2023) 107149.

[114] M. Gezimati, G. Singh, Forecasting healthcare 5.0 driven internet of medical things for a seamless continuum of care, IEEE Access (2025).

[115] M. Joshi, M. K. Mishra, S. Karthikeyan, Impact of hardware connectivity on grover's algorithm in nisq era, Quantum Information Processing 24 (4) (2025) 1–21.

[116] B. Sharma, D. Kaushal, M. Sharma, S. Joshi, S. Gopal, P. Gupta, Integration of ai, digital twin and internet of medical things (iomt) for healthcare 5.0: a bibliometric analysis, in: 2023 International Conference on Advances in Computation, Communication and Information Technology (ICAICCIT), IEEE, 2023, pp. 1313–1318.

[117] J. Dutta, D. Puthal, Advancing ehealth in society 5.0: a fuzzy logic and blockchain-enhanced framework for integrating iomt, edge, and cloud with ai, IEEE Access (2024).

[118] H. Kurniawan, L. Rodríguez-Soriano, D. Cuomo, C. G. Almudever, F. G. Herrero, On the use of calibration data in error-aware compilation techniques for nisq devices, in: 2024 IEEE International Conference on Quantum Computing and Engineering (QCE), Vol. 1, IEEE, 2024, pp. 338–348.

[119] J.-B. Waring, C. Pere, S. Le Beux, Noise aware utility optimization of nisq devices, in: 2024 22nd IEEE Interregional NEWCAS Conference (NEWCAS), IEEE, 2024, pp. 168–172.

[120] X.-D. Xie, X. Zhang, B. Koczor, X. Yuan, Advances in quantum computation in nisq era (2025).

[121] Y. Mohamed, A. Elghadban, H. I. Lei, A. A. Shih, P.-H. Lee, Quantum machine learning regression optimisation for full-scale sewage sludge anaerobic digestion, npj Clean Water 8 (1) (2025) 17.

[122] M. Nadim, M. Hassan, A. K. Mandal, C. K. Roy, Quantum vs. classical machine learning algorithms for software defect prediction: Challenges and opportunities, in: 2025 IEEE/ACM International Workshop on Quantum Software Engineering (Q-SE), IEEE, 2025, pp. 35–42.

[123] S. Ashwani, A. J. Tripathy, S. Karna, P. R. Jahanve, S. M. Rajagopal, Quantum computing for climate change: A comprehensive review of current applications, challenges, and future directions, in: 2024 15th international conference on computing communication and networking technologies (ICCCNT), IEEE, 2024, pp. 1–7.

[124] A. Shehata, T. Naughton, I.-S. Suh, A framework for integrating quantum simulation and high performance computing, in: 2024 IEEE International Conference on Quantum Computing and Engineering (QCE), Vol. 2, IEEE, 2024, pp. 300–305.

[125] A. A. Gentile, S. Acheche, A. Ghosh, O. Kyriienko, L.-P. Henry, Hardware-informed applications, application-informed hardware, IEEE Nanotechnology Magazine (2025).

[126] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, H. Neven, Barren plateaus in quantum neural network training landscapes, Nature communications 9 (1) (2018) 4812.

[127] E. Grant, L. Wossnig, M. Ostaszewski, M. Benedetti, An initialization strategy for addressing barren plateaus in parametrized quantum circuits, Quantum 3 (2019) 214.

[128] A. Skolik, J. R. McClean, M. Mohseni, P. Van Der Smagt, M. Leib, Layerwise learning for quantum neural networks, Quantum Machine Intelligence 3 (1) (2021) 5.

[129] H.-X. Tao, R.-B. Wu, On the role of controllability in pulse-based quantum machine learning models, in: 2024 IEEE 63rd Conference on Decision and Control (CDC), IEEE, 2024, pp. 13–18.

[130] S. Joshi, D. Justice, U. Stege, piqture: A qml library for image processing, in: 2024 IEEE International Conference on Quantum Computing and Engineering (QCE), Vol. 2, IEEE, 2024, pp. 583–584.

[131] M. A. Haque, V. Kumar, S. N. Khan, J.-J. Kim, et al., Quantum intelligence in drug discovery: Advancing insights with quantum machine learning, Drug Discovery Today 30 (10) (2025) 104463.

[132] M. A. Khan, M. N. Aman, B. Sikdar, Beyond bits: A review of quantum embedding techniques for efficient information processing, IEEE access 12 (2024) 46118–46137.

[133] A. Aggarwal, S. V. Singh, S. Bansal, V. Bhutani, A detailed overview of quantum computing machine learning techniques, in: 2024 International Conference on Communication, Computer Sciences and Engineering (IC3SE), IEEE, 2024, pp. 1721–1725.

[134] A. C. Dhanu, P. Chaitra, M. Manohar, A comparative analysis of classical and quantum machine learning approaches in healthcare data analytics, in: Machine Intelligence and Digital Interaction Conference, Springer, 2024, pp. 83–96.

[135] S. Pandey, S. Pandey, B. Bhushan, P. Baniya, A. Agarwal, Utilizing graph neural networks (gnn) in quantum-natural language processing (q-nlp) for risk management in banking sector: A novel approach, in: Graph Neural Networks: Essentials and Use Cases, Springer, 2025, pp. 109–134.

[136] S. Gholampour, Impact of nature of medical data on machine and deep learning for imbalanced datasets: clinical validity of smote is questionable, Machine Learning and Knowledge Extraction 6 (2) (2024) 827–841.

[137] D. Herman, C. Googin, X. Liu, Y. Sun, A. Galda, I. Safro, M. Pistoia, Y. Alexeev, Quantum computing for finance, Nature Reviews Physics 5 (8) (2023) 450–465.

[138] P. Steinmüller, T. Schulz, F. Graf, D. Herr, explainable ai for quantum machine learning, 2022.

[139] X. Wang, Y. Wang, B. Qi, R. Wu, Limitations of amplitude encoding on quantum classification, arXiv preprint arXiv:2503.01545 (2025).

[140] D. Ranga, A. Rana, S. Prajapat, P. Kumar, K. Kumar, A. V. Vasilakos, Quantum machine learning: Exploring the role of data encoding techniques, challenges, and future directions, Mathematics 12 (21) (2024) 3318.

[141] L. Wei, H. Liu, J. Xu, L. Shi, Z. Shan, B. Zhao, Y. Gao, Quantum machine learning in medical image analysis: A survey, Neurocomputing 525 (2023) 42–53.

[142] S. Raza, A. Shaban-Nejad, E. Dolatabadi, H. Mamiya, Exploring bias and prediction metrics to characterise the fairness of machine learning for equity-centered public health decision-making: A narrative review, IEEE Access (2024).

[143] S. Lambiase, M. De Stefano, F. Palomba, F. Ferrucci, A. De Lucia, Socio-technical well-being of quantum software communities: An overview on community smells, in: Euromicro Conference on Software Engineering and Advanced Applications, Springer, 2025, pp. 39–56.

[144] J. Seol, J. Kim, A. Kancharla, Integrating blockchain-based security and privacy with qml in edge computing for 6g networks, in: International Conference on Computer Applications in Industry and Engineering, Springer, 2024, pp. 86–101.

[145] H. Singh, Managing the quantum cybersecurity threat: Harvest now, decrypt later, in: Quantum Computing, CRC Press, 2024, pp. 142–158.

[146] A. Sonavane, A. Aylani, Exploring the use of quantum algorithms, Quantum Computing and Artificial Intelligence in Logistics and Supply Chain Management (2025).

[147] P. K. Donta, A. Saleh, Y. Li, S. Vaishnav, K. Fang, H. Feng, Y. Xia, T. R. Gadekallu, Q. Zhang, X. Shi, A. Beikmohammadi, S. Magnússon, I. Murturi, C. K. Dehury, M. Paprzycki, L. Loven, S. Tarkoma, S. Dustdar, Socio-technical aspects of agentic ai (2025). arXiv:2601.06064.